

Leisure-Time Physical Activity in School Environments: An Observational Study Using SOPLAY

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Background. Schools provide settings for physical activity (PA), but objective tools for measuring PA are lacking. We assessed an instrument to directly observe group PA and measured the leisure-time PA of adolescents throughout the school day.

Methods. Leisure-time PA was studied by direct observation in 24 middle schools in Southern California using SOPLAY (System for Observing Play and Leisure Activity in Youth). SOPLAY uses momentary time sampling to record the activity of each individual as sedentary, walking, or very active.

Results. The largest proportion of students visited activity areas at lunch time (19.5%), followed by before (4.1%) and after school (2.1%). More boys than girls visited activity areas before school (33.4 vs 7.7) and at lunch time (148.8 vs 36.7). Boys in areas engaged in more moderate to vigorous PA than girls before school (59.6 vs 40%) and at lunch time (67.7 vs 51.7%).

Conclusions. Few students use opportunities to be physically active during lesiure time at school. Policies and environmental manipulations (e.g., supervision, equipment, structured programs) are needed to attract more adolescents, especially girls, to existing activity areas. The feasibility of SOPLAY for measuring group PA was established. © 2000 American Health Foundation and Academic Press

Key Words: adolescents; middle school; physical activity; measurement; exercise; leisure activity (recreation).

INTRODUCTION

Regular physical activity (PA) is an important part of a healthful lifestyle for children and adolescents, and national and international guidelines for youth activity

levels have been developed [1–3]. Many adolescents are not meeting these guidelines, particularly for high-intensity physical activity [4]. Numerous obstacles impede youth from obtaining recommended levels of physical activity, and, as they grow older, their activity levels decline [5]. Children and adolescents obtain most of their physical activity in nonschool settings [6]. Many, however, have limited access to structured activity environments such as youth sports and activity lessons and to nonstructured environments such as parks and hiking trails. Girls are particularly disadvantaged by fewer opportunities for youth sports [7] and the less affluent are hampered by entry fees and transportation costs. Adolescents, too young to drive, also must rely on others to provide them access to distant activity sites.

Schools have been identified as providing the most logical environment for promoting public health of youth through physical activity [8,9]. Most studies of children's physical activity have taken place in elementary schools, primarily during physical education classes [10]. However, there are nearly 14,000 middle and junior high schools in the United States [11] and the physical activity that adolescents accrue in these environments has rarely been studied, particularly activity obtained during leisure-time periods. Almost all middle schools have facilities to support leisure-time physical activity during the day, and many conduct structured extracurricular activities, including intramural and interscholastic sports and club programs.

There are several reasons for interest in studying youth physical activity at school, in addition to physical education. First, even the best school physical education programs do not provide enough physical activity to meet health-related recommendations [9–13]. Thus, increasing physical activity at other times throughout the school day could help young people achieve recommended levels. Second, the recent, rapid increase in youth obesity [14] is an important stimulus for increased physical activity in numerous settings. Third,

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keeping students engaged in physical activity on campus after school has the potential to decrease delinquency, risky health behaviors, and social problems that are most likely to occur in the after-school hours. Prior to considering interventions, however, it is important to describe current activity levels throughout the school day.

Numerous objective methods have been designed and validated for measuring the physical activity of *individuals*, including direct observation, heart rate monitors, accelerometers, and doubly labeled water [15], but no validated instruments to assess the physical activity of groups could be located. This lack of an objective tool has hampered investigations of physical activity in “open” environments, such as recreational and leisure settings. Measuring activity in these settings is complicated because both the number of participants and their activity levels change frequently. Because leisure time activity may occur in diverse locations, it is also desirable to assess relevant environmental aspects of the settings that might influence physical activity.

There were two purposes of the present study. The primary purpose was to investigate the physical activity levels of boys and girls and associated environmental characteristics before school, at lunch time, and after school in 24 middle schools over a 20-week period. To meet this objective, a direct observation system (System for Observing Play and Leisure Activity in Youth—SOPLAY) was developed. The secondary purpose was to evaluate SOPLAY as an instrument for measuring physical activity in leisure and sport settings.

METHODS

Schools

Middle schools (grades 6–8) within a 1-h drive of a university campus in Southern California were recruited to participate in an intervention study targeting physical activity and nutrition. Intervention and measurement procedures were approved by the university human subjects review board, and the first 24 schools to sign a participation agreement were accepted into the study. The study sample included 24 schools from nine districts, with a range of 1–6 schools per district. Schools had a mean enrollment of 1081 (SD = 352; median = 1004) students. Averaged across schools, 49% of students were female, 43% (SD = 19) were nonwhite, 39% (SD = 21) received free or low-cost meals, and 38% (SD = 26) were bused to school. Median start time for classes was 7:45 AM (range = 7:12 AM to 9:00 AM) and median end time was 2:20 PM (range = 2:05 PM to 3:35 PM). On average, schools held two lunch periods per day (range = 1–3) lasting 35.5 (SD = 6.8) min.

Target Areas

All potential areas for leisure-time physical activity in each school were identified and measured prior to

data collection. Agreement among assessors was established on the location, size, and boundaries of each target area, and maps detailing target areas for each school were made. A total of 151 areas were targeted for observation, ranging from 2 to 8 (mean = 6.3) per school. The areas included 3 swimming pools, 3 weight rooms, 7 gymnasiums, 18 multipurpose rooms, 1 auditorium, 60 outdoor marked court spaces, and 60 outdoor play areas (32 grassy fields and 28 areas with a nongrass surface).

Observation Instrument

SOPLAY. SOPLAY was designed to obtain observational data on the number of participants and their physical activity levels during play and leisure opportunities. The system is based on momentary time sampling [16] and uses Placheck recording (i.e., Planned Activity Check) which involves group time sampling techniques [17]. Placheck has previously been used to examine children's aggressive behavior on playgrounds [18].

Systematic SOPLAY scans of target areas were made during three measurement periods (before and after school and during lunch time). During a scan the physical activity of each individual in a target area was coded as sedentary (i.e., lying down, sitting, or standing), walking, or very active. These activity codes have been validated by heart rate monitoring [19,20] and permit energy expenditure rates (EER) to be estimated. Separate scans were made for girls and boys. Simultaneous entries were made for the time of the observation and for contextual characteristics of each area such as its accessibility (e.g., not locked), usability, and whether supervision, organized activities, and equipment were provided. The predominant type of activity in each area that girls and boys were engaging in was also recorded, using a list of 14 activities (e.g., basketball, football). A description of the SOPLAY codes and the measurement protocol is available from the lead author.

Activity variables. For activity measures, counts were tallied for those engaged in sedentary, walking, and very active behavior in each area to obtain a summary score. The counts were transformed into estimates of energy expenditure rate (kcal/kg/min) separately for the girls and boys. To estimate kcal/kg/min expended, the numbers counted in the sedentary, walking, and very active categories were multiplied by the constants 0.051, 0.096, and 0.144 kcal/kg/min, respectively. These kcal/kg/min estimates were derived from a previous study [19]. A second summary score, moderate to vigorous physical activity (MVPA), was created by summing the walking and very active categories.

Observation schedule and protocol. Systematic observations were made periodically in each area during

three randomly scheduled days at each school over 20 weeks (total = 72 days). Each activity area was scanned visually from left to right and the code representing each student's activity level was entered into a handheld mechanical recording device at an approximate rate of one child per s. Observations were conducted in all activity areas before school, at lunch time, and after school. All activity areas were observed in a specific order for each observation period. Before school observations were staggered every 5 minutes so that the last area was observed 15 minutes prior to the start of school. Lunch time observations began 15 minutes into the lunch period. After-school observations were made at 15, 45, and 75 minutes after classes were dismissed for the day.

Observer training. Four certified SOPLAY assessors conducted all observations. Training consisted of a 3-day workshop led by the primary author. The first day (6 h) included classroom lectures designed to familiarize trainees with operational definitions, activity level discrimination, instrument notation, and coding conventions. The second day of training (6 h) began with a review of definitions followed by brief trials of coding using videotaped samples of adolescents' free play. This was followed by an assessment of trainees using precoded behavioral vignettes on videotape. These served as "gold-standard" observations. All trainees were required to meet acceptable criteria for reliable assessment (observer agreement with gold-standard >80%; intraclass correlations >0.9 for energy expenditure estimates) before progressing to the final stage of training. The final day of training (8 h) was composed of field-based observations using the SOPLAY instrument in "real-time." Observers coded samples of free-play activity in middle school environments (low student density/low activity, high student density/low activity, low student density/high activity, and high student density/high activity).

Recalibration. During week 9 of the data collection period (20 weeks), a retraining and assessment session was conducted. Feedback from videotaped assessments was provided to counter observer drift from the "gold-standard" and reduce any interobserver disagreement.

Reliability. Reliability data were collected during 14 of the 72 measurement days (20%) by rotating pairs of assessors who made simultaneous and independent observations. The 14 days were scheduled periodically throughout the 20-week measurement period and each reliability day was at a different school. Data from observations in 185 activity areas were used in reliability analysis.

To examine the reliability of activity counts made by different assessors a series of paired *t* tests and intraclass correlations were computed. Testing mean

counts of students by gender and activity level revealed no significant differences between independent assessors, except for counts of very active boys (1.72 vs 2.01 boys; $t(184) = 2.06, P < 0.05$). Energy expenditure rate estimates also revealed significant differences for very active boys (0.075 vs 0.079 kcal/kg/min/area; $t(184) = 2.24, P < 0.05$). Interobserver intraclass correlations were high for counts of sedentary ($R = 0.98$) and walking girls (0.95), although lower for very active girls (0.76). The correlation for very active girls was lower because of the low frequency of occurrence. For boys, correlations were high for sedentary (0.98), walking (0.98), and very active (0.97) behavior. For estimates of energy expenditure, intraclass correlations were high (0.99 for both boys and girls).

Interobserver agreements for the five contextual variables were high: 95% for area accessibility, 97% for usability, 93% for presence of supervision, 96% for presence of organized activity, and 88% for provision of equipment. It was concluded that all interobserver agreements and intraclass correlations met acceptable criteria (i.e., interobserver agreement = 80% and $R = 0.75$) for reliable assessment. Across 3 measurement days, the intraclass correlation of the SOPLAY measure for estimating energy expenditure was 0.88 and 0.75 for boys and girls, respectively. The intraclass correlation for boys suggests that 3 days of measurement provides an adequate sampling for a reliable ($r = >0.8$) SOPLAY measure. Using the Spearman-Brown prophecy formula [21], 4 measurement days would be necessary to reach a reliability level of 0.8 when measuring girls.

Validity. The SOPLAY activity codes have been used in previous observation systems: BEACHES (Behaviors of Eating and Activity for Children's Health: Evaluation System) [19] and SOFIT (System for Observing Fitness Instruction Time) [22]. Construct validity of the activity codes has been established through heart rate monitoring [19,20]. There is also an abundance of evidence demonstrating that momentary time sampling techniques yield valid behavioral samples [16].

The number of students observed in moderate to vigorous physical activity before school, at lunch, and after school was correlated with self-reported levels of activity (minutes per week) derived from a student survey distributed to each school during the same semester. Surveys were randomly distributed to three classes at each school (one per grade level) and included a small financial incentive for completion. A total of 1678 surveys were returned, yielding a return rate of 72%. Using mean scores per school ($n = 24$), Pearson correlations of observed MVPA and self-reported physical activity before school, at lunch time, and after school were 0.71, 0.73, and 0.35, respectively. These findings provide initial support for the concurrent validity of SOPLAY. The

lower correlation observed for after-school activity is probably more attributable to measurement error in the self-report instrument than error in SOPLAY observations. In addition, the very low number of children participating after school produced a restricted range that suppresses correlations.

Data Analysis

Units of analysis. For analyses involving contextual characteristics of activity areas, the unit of analysis was one scan of one area. For all other variables the unit of analysis was a gender- and period-specific daily summary for each school, with the exception of summary school-level variables (e.g., total play space per school), where school was the unit of analysis. To compute gender- and period-specific daily summaries for each school, daily means of activity counts based on the 3 observation days for each target area at each school were computed. Using these scores, outcome variables were computed by aggregating counts across activity areas separately by gender and period for each school. For some statistical analyses, a mean of the three after-school observations was used to provide a single measure of after school activity.

Dependent variables. For each contextual characteristic, percentage of occurrence (e.g., the proportion of scans in which areas were accessible to students) was computed based upon the total number of scans across all activity areas. The percentage of occurrence measures were used as dependent variables when comparing area contexts across the school day. For activity variables, counts of boys and girls were divided by the gender-specific daily attendance to derive the percentage of school attendance present in activity areas at any particular time. The proportion of school attendance engaged in MVPA (i.e., number of students coded as walking plus very active divided by the daily school attendance) and estimates of energy expenditure rate (described earlier) were included as dependent variables. Activity type (expressed as proportion of students engaged in selected activities) was also included as a dependent variable. Low participation in specific sports and games by both boys and girls, however, necessitated collapsing activities into three categories: unstructured/unspecified activity, structured sports, and structured playground games.

Independent variables. Gender (two levels) and observation period (five levels; before school, lunch time, and three after school) were the primary independent variables used for analyses.

Statistical inference. Analyses involving percentage of occurrence (i.e., frequency) data were analyzed using χ^2 . To analyze gender and period differences in activity

variables, a series of 2×5 (gender by period) mixed-model ANCOVAs were computed using the PROC MIXED procedure in SAS (SAS Institute Inc., 1996). Each mixed model was fitted to the data using the restricted maximum likelihood method and assumes a compound symmetry covariance structure. For each mixed model, gender and period were included as fixed effects, with school included as a random effect to account for the effects of clustering by school. The percentage of students bused to each school was included as a covariate because prior analyses indicated that busing was related to on-campus physical activity before and after school. Tukey post hoc comparisons were computed for significant period effects. All means and standard errors were adjusted for covariate and clustering effects, unless stated otherwise. Pearson correlations were used to examine the association between school characteristics and school-level activity variables.

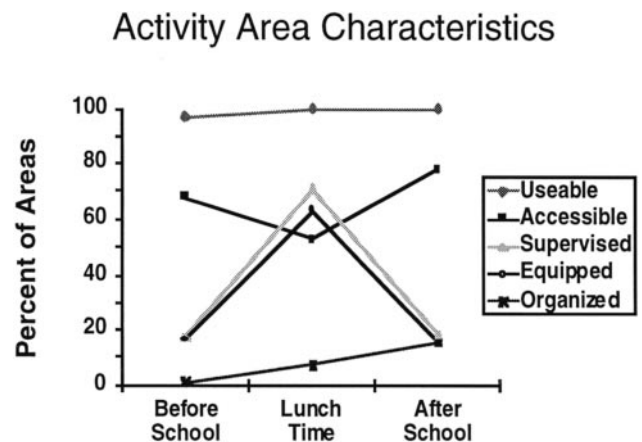
RESULTS

Physical Activity Areas

Schools had an average of 6.3 (SD = 1.6) separate areas for physical activity to take place. The mean size of areas was 6921 (SD = 10,035) square meters, yielding, on average, a total of 43,545 (SD = 19,211) square meters of available play space per school. The Pearson correlation of number of areas per school with total available play space was 0.25.

Contextual Variables

Figure 1 shows the percentage of activity areas that were usable, accessible to students, and had supervision, equipment, and organized activities available during the three time periods. The areas were nearly always usable, but were accessible to students only about



(N=24 middle schools; 151 activity areas; 3 days each)

FIG. 1. Characteristics of observed physical activity areas during three daily time periods.

66% of the time. Areas were more accessible after school than during other time periods [χ^2 (2, N = 2544) = 141.09, P < 0.001]. They were supervised [χ^2 (2, N = 1710) = 455.43, P < 0.001] and had equipment provided [χ^2 (2, N = 1710) = 403.87, P < 0.001] more during lunch time than before and after school. Organized activities occurred infrequently (mean = 8% of the observations) and rarely happened before school.

Activity Counts

Table 1 displays the mean number of students (SD) and adjusted percentage of daily attendance, percentage engaged in MVPA, and energy expenditure estimates, by gender, in the activity areas before school, at lunch time, and after school. Results of the mixed-model ANCOVAs (adjusting for the percentage of students bused to school and clustering by school) for the percentage of daily attendance, percentage of students engaged in MVPA, and energy expenditure estimates, by gender and measurement period, are presented in Table 2.

There were both time (i.e., daily measurement period) and gender differences in area usage. Significantly more boys than girls visited activity areas at lunch time (30.6 vs 8.3% of daily attendance, P < 0.001), but not before (6.7 vs 1.6%) or after school (1.9 vs 2.2%). Both boys and girls were more likely to visit activity areas at lunch time than before (P < 0.001) and after school (P < 0.001). A significant gender by period interaction (P < 0.001) indicated that lunch time had a more powerful influence on attracting boys to activity areas than girls.

Physical Activity Levels.

Physical activity levels of those in activity areas differed by gender. Overall (i.e., across measurement periods) boys in areas engaged in more MVPA than girls (adjusted grand means = 62.4 vs 48.3% in MVPA) and had a higher average energy expenditure rate (0.083 vs 0.070 kcal/kg/min, P < 0.004). The energy expenditure rate and percentage in MVPA of those in activity areas not differ significantly across observation periods.

School Trends

Pearson correlations (data not shown) revealed that the percentage of school attendance visiting activity areas was not significantly associated with total activity space available, proportion of students who were nonwhite, or proportion receiving free or reduced meals. However, there was a significant positive relationship (r = 0.40, P < 0.05) between the percentage of daily school attendance present in activity areas and the proportion of students bused to school. No significant associations were evidenced between energy expenditure variables and selected school characteristics.

Types of Activity

The most prominent activity that both girls and boys engaged in was coded as “no identifiable sport, game, or exercise.” For girls, this was coded 94% of the time before school, 73% at lunch time, and 87% after school. For boys, these activities were coded 70% of the time before school, 37% at lunch time, and 84% after school. Activities with more structure (i.e., sports and games)

TABLE 1

Number of Students (Mean, SD), Adjusted Percentage of Daily School Attendance, Percentage in Moderate to Vigorous Physical Activity (MVPA), and Energy Expenditure Rate Estimates (EER), by Gender, in All On-Campus Activity Areas before School, at Lunch Time, and after School (N = 24 Schools)

	Before school	Lunch time	After school	Post hoc
All students				
Number (SD)	41.1 (43.5)	185.5 (121.1)	20.3 (16.6)	—
% of attendance	4.1	19.5	2.1	L > BS, AS*
% in MVPA	51.2	59.3	56.2	ns
EER (kcal/kg/min)	0.070	0.082	0.078	ns
Girls				
Number (SD)	7.5 (12.1)	36.7 (35.2)	10.2 (10.2)	—
% of attendance	1.6	8.3	2.2	L > BS, AS*
% in MVPA	40.0	51.7	53.1	ns
EER (kcal/kg/min)	0.057	0.075	0.078	ns
Boys				
Number (SD)	33.4 (33.3)	148.8 (91.3)	10.0 (10.0)	—
% of attendance	6.7	30.6	1.9	L > BS, AS*
% in MVPA	59.6	67.7	59.9	ns
EER (kcal/kg/min)	0.082	0.088	0.079	ns

Note. BS, before school; L, lunch time; AS, after school.

* P < .001.

TABLE 2

Results of Mixed-Model ANCOVAs (Adjusting for Clustering by School and Percentage of Students Bused to Campus) for the Percentage of Daily Attendance, Percentage of Students Engaged in Moderate to Vigorous Physical Activity (MVPA), and Energy Expenditure Estimates (EER), by Gender and Period (before and after School and during Lunch Time)

	<i>F</i> statistic	ndf, ddf	<i>P</i>
Gender (main effect)			
% of attendance	108.14	(1, 211)	<0.001
% in MVPA	16.88	(1, 185)	<0.001
EER (kcal/kg/min)	8.32	(1, 211)	<0.004
Period (main effect)			
% of attendance	159.55	(2, 211)	<0.001
% in MVPA	2.24	(2, 185)	<0.12
EER (kcal/kg/min)	2.10	(1, 211)	<0.13
Gender * period (interaction)			
% of attendance	65.80	(2, 211)	<0.001
% in MVPA	1.63	(2, 185)	<0.20
EER (kcal/kg/min)	2.64	(1, 211)	<0.07
% bused (covariate)			
% of attendance	3.59	(1, 211)	<0.06
% in MVPA	2.58	(1, 185)	<0.11
EER (kcal/kg/min)	0.30	(1, 211)	<0.58

occurred mostly at lunch time. Basketball was the mostly frequent occurring sport (13% of the time for boys and 5.7% for girls).

DISCUSSION

This is the first large study to assess the physical activity of adolescents on school campuses during leisure-time periods. In addition to physical education classes, there are numerous opportunities for adolescents to be physically active at school. All 24 middle schools had areas that were appropriate for PA and were available before and after school and at lunch time. Although students were allowed to use these activity areas, the results indicate that very few students visited them during leisure-time periods (ranging from 2–4% of daily attendance before and after school to 20% during lunch time). Students who did visit areas, however, were quite active, with 40–68% being physically active. The activities they engaged in were predominately without an identifiable structure, so most students “hung out,” rather than engaging in a specific sport, game, or exercise.

The results suggest that numerous opportunities exist for increasing the physical activity levels of students during time periods when they could choose to be active or inactive. The provision of more supervision, equipment, and organized activities during these times might lead to more students being more physically active. Because a large proportion of adolescents are not meeting PA guidelines [4], it is worthwhile to examine methods of enhancing the school environment to increase

physical activity throughout the school day, because virtually all middle school students could be reached through such interventions.

Target areas were physically usable most of the time; they frequently were not accessible to students for physical activity (34% of the time). School policy and lack of supervision often were cited as reasons why students were not permitted in areas. Additionally, there was little apparent effort to promote physical activity on campus before and after school. Few organized physical activities were observed during these periods, and areas were seldom supervised or had equipment accessible. Additionally, middle school students are not old enough to drive so transportation problems may have been a factor in limiting the numbers that used activity areas before and after school. Altering bus schedules might help to overcome this problem.

Of the three observation periods, the lunch period was the most popular time for physical activity. All students were required to remain on campus during lunch time, and this was when supervision and equipment were provided most often.

Gender differences in on-campus physical activity were pronounced. Compared to girls, boys visited targeted areas more often, were more physically active, and participated in activities that had more structure. Attention must be paid to programming middle school environments to make activity opportunities more attractive to all students, but girls should receive special consideration. Particular focus should be given to identifying activity preferences and recruitment strategies that can be used to make activities at school more appealing to girls. It is important to ensure that potentially aversive events associated with engaging in physical activity are reduced as much as possible (e.g., by providing sufficient time and privacy for clothing changes).

This study used a highly reliable direct observation instrument to assess on-campus physical activity during three leisure-time periods. It complements a concurrent study of student physical activity, lesson context, and teacher behavior during structured physical education classes in the same schools [13]. Activities that students engaged in off campus were not observed, and accurate estimations for the duration of activity were not possible using momentary time sampling. While all likely areas for physical activity in each school were targeted for observation, it is possible for physical activity to have occurred in an unlikely space on some occasions. Another limitation is that SOPLAY was designed to assess the physical activity participation of a group in a specific setting and does not permit evaluations of individuals.

The study was conducted in 24 middle schools in a

large culturally and geographically diverse metropolitan area over an extended time period (20 weeks). Several factors must be considered when attempting to generalize the results. Observations took place from January to June and student physical activity during the fall might yield different results. Additionally, all observations were in Southern California and there may be environmental and political factors that restrict the generalizability of present findings. For example, a temperate climate permits physical activity to occur outdoors year-round; thus, schools in other locations may have more indoor and fewer outdoor activity spaces.

Regarding the secondary purpose of the study, the feasibility of SOPLAY was established through 72 days of measurement at 24 diverse schools. Interobserver reliabilities for counts of boys and girls in targeted activity areas, their physical activity levels, and all coded contextual variables were very high. The number of students observed in school activity areas was significantly correlated with the number of students who self-reported that they typically used school activity areas. The activity codes were supported by heart rate monitoring [19,20] which permitted energy expenditure rates to be estimated. A limitation of SOPLAY is the broad categories of activity level, which do not permit detailed analyses of activity intensities. Although a wide range of intensities are covered by the code "very active," early development work indicated that finer discriminations were not reliably coded. However, the instrument is useful for researchers and practitioners interested in assessing the numbers of participants and their physical activity levels in leisure settings. Comparisons can be made among different environments or within the same environment over time. Although the present study evaluated SOPLAY on middle school campuses, the instrument was designed to be used in a range of settings such as parks, playgrounds, and other indoor and outdoor environments. A strength of SOPLAY is that simultaneous evaluations of physical activity and numerous contextual factors in the social and physical environment allow for examining the influence of setting variables on attendance and physical activity levels. It has the potential to assess the effects of environmental and policy interventions that may be implemented to influence the physical activity of entire populations [23].

The primary purpose of the study was to investigate on-campus physical activity levels throughout the school day as well as to document relevant school environment characteristics. Before school, after lunch, and after school were defined in advance as time periods when students had opportunities to be physically active. The main findings were that: (a) although physical settings were available for physical activity; (b) school

officials made limited efforts to provide equipment, supervision, or structured activities to support and encourage student physical activity; and (c) relatively few students sought out opportunities for physical activity during leisure periods. It is recognized that not all students can be on campus before and after school, but the extensive school facilities received little use, because only 2–4% of students were observed in the activity areas. Similarly, not all students can be expected to use their free time for physical activity after eating lunch, but only 30% of boys and 8% of girls were observed in activity areas during this time period, even though all students were required to remain on campus. The authors interpret these results as indicating that many opportunities for physical activity by middle students are being wasted. The large gender differences suggest that special efforts may be required to make activity areas and physical activities during leisure time at school more attractive to girls. The pressing need for interventions to increase leisure-time physical activity at multiple settings including schools is demonstrated by the rising prevalence of youth obesity [14], the multiple health problems associated with physical inactivity during youth [1], and the high proportion of young people who do not meet health-related physical activity guidelines [1,4].

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